

FIELD VISIT PHOTOGRAPHS

Image.1

The prototype device is mounted at the boarding edge of the BEST Route 326 stop during field testing at Dindoshi Bus Depot. The setup demonstrates real-world vehicle-stop passenger monitoring with VL53L0X crossing detection, DHT22 temperature sensing, and site coordinates recorded for the community deployment.



Image.1: Field Demonstration

Image.2

Close-up view showing the enclosed electronic components mounted securely at the stop for sensor-based overcrowding monitoring: ESP32 DoIT DevKit V1, VL53L0X (I2C GPIO21/GPIO22), DHT22 (GPIO4), and red/green LEDs (GPIO25/GPIO26) on the 3.3 V rail.



Image.2: Close-up Prototype at the Bus Stop

Image.3

The ESP32 Serial Monitor displays ToF distance, consecutive-hit confirmation, DHT22 temperature and humidity, passenger count on V1, minutes since last crossing on V8, Python risk on V3, and LED status. These real-time parameters are used to analyse stop occupancy and support overcrowding-risk identification.

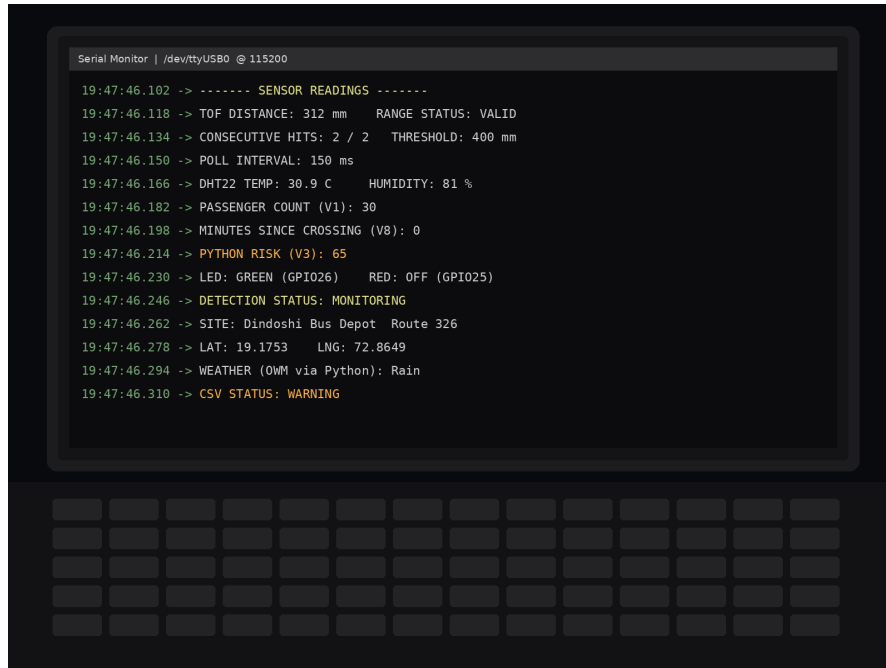


Image.3: Real-Time Sensor Data Monitoring

Image.4

This image shows the Dindoshi / Route 326 boarding edge after rainfall. Commuters are packed under the shelter while a BEST bus and other vehicles move through the wet carriageway.



Image.4: Field Observation of an Overcrowded Bus Stop

Rain makes boarding slower and the waiting area more congested. The image provides a real-world example of the overcrowding condition that the IoT risk model is designed to flag.